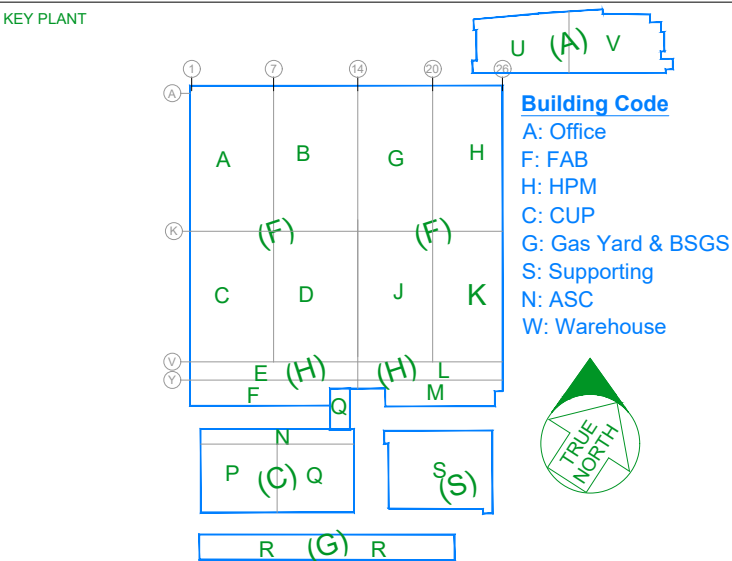
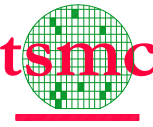


HVAC PUMP SCHEDULE															
UNIT MARK	SERVICE	TYPE	SUCTION SIZE A	DISCHARGE SIZE A	FLOW L/S	HEAD M	BHP	RPM	MOTOR			FLUID	REMARKS		
									KW	VOLTAGE	TYPE				
22P1-C10P-5CH3-P01	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	190	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1) (STANDBY)		
22P1-C10P-5CH3-P02	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-5CH3-P03	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P04	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P05	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P06	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P07	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P08	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P09	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-5CH3-P10	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)		
22P1-C10Q-5CH3-P11	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)		
22P1-C10Q-5CH3-P12	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)		
22P1-C10Q-5CH3-P13	CHILLED WATER PRODUCTION	IN-LINE	300	300	272	42	-	-	150	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE , SPACE)		
22P1-C10P-12CH3-P01	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P02	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P03	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P04	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P05	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P06	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P07	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10P-12CH3-P08	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P09	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P10	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P11	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P12	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P13	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P14	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (STAGE 1)		
22P1-C10Q-12CH3-P15	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)		
22P1-C10Q-12CH3-P16	CHILLED WATER PRODUCTION	IN-LINE	300	300	400	44	-	-	223.5	3/460/60	TEFC				



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Taiwan Semiconductor Manufacturing Company, Ltd.

ACCEPTANCE REVIEWED	DATE	ACCEPTANCE REVIEWED	DATE
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CONTRACTOR

uis 漢唐集成股份有限公司
United Integrated Services

DRAWN	TCM	DATE 2023/11/03	RECHECKED Steven Yu	DATE 2023/11/03
DESIGNED	TCM	DATE 2023/11/03	APPROVED JOSEPH LEE	DATE 2023/11/03

MECHANICAL SYSTEM EQUIPMENT SCHEDULE (2)

PROJECT FAB 22 PHASE 1 PROJECT

PACKAGE NAME	MEP SYSTEM
DRAWING NAME	MEP SYSTEM

F22P1-M-1-C-02

SCALE	UNIT
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NOT TO SCALE

FILE NAME	TEST.M: 0.00	Rev
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TSMC Confidential Information



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TITLE

MECHANICAL SYSTEM
EQUIPMENT SCHEDULE
(1)

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PROJECT	FAB 22 PHASE 1 PROJECT	
PACKAGE NAME	MEP SYSTEM	
DRAWING NAME	F22P1-M-1-C-01	
SCALE	NOT TO SCALE	UNIT mm
FILE NAME	F22P1-M-1-C-01	Rev. A

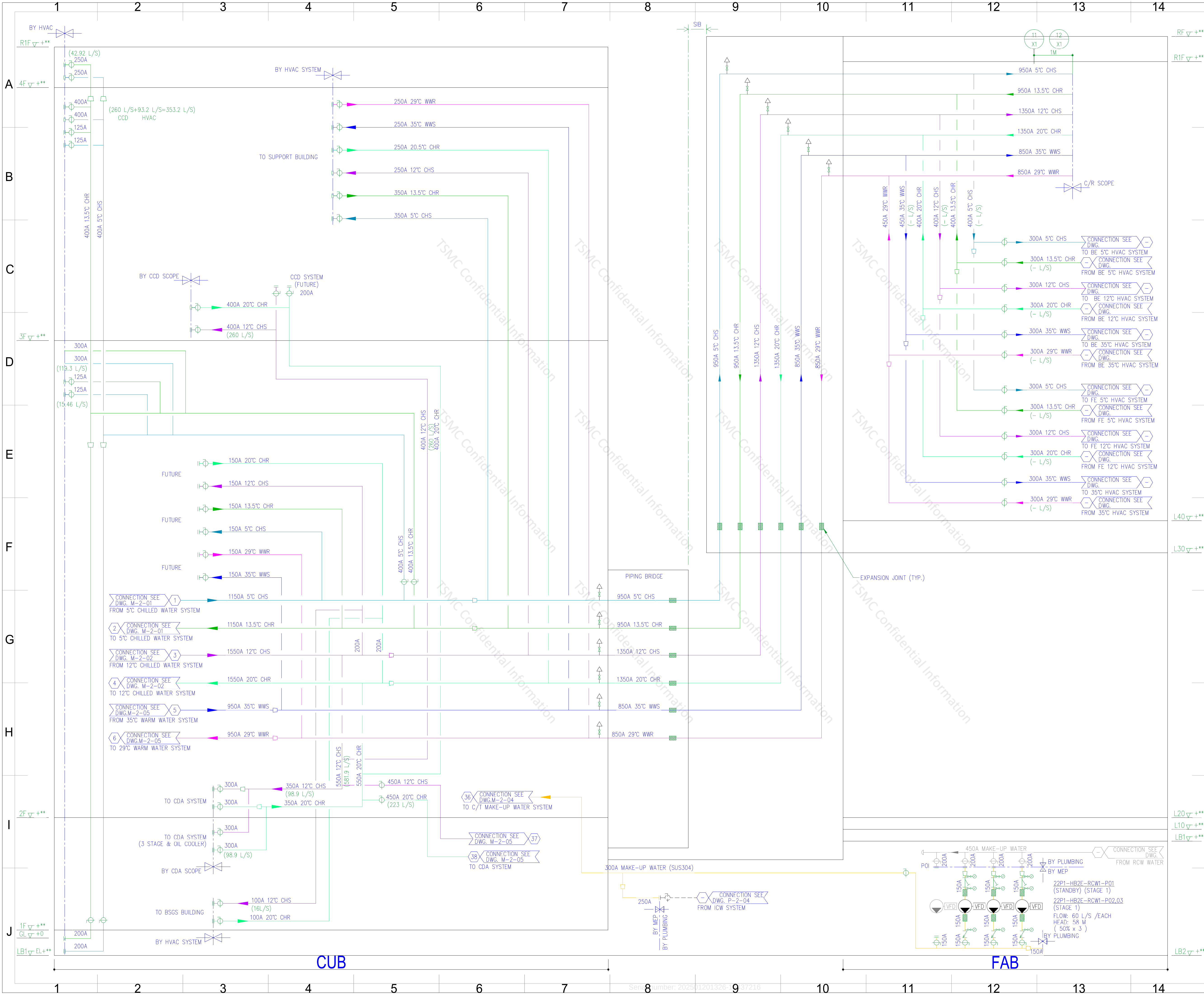
UPW/DI BUFFER TANK SCHEDULE						
UNIT MARK	SERVICE	QTY	MATERIAL	EFFECTIVE TANK CAPACITY (M ³)	DIMENSION (mm)	REMARKS
22P1-FB2H-DIW2-T01	22P1B UPW/DI MAKE UP TANK	1	SUS 316	15		
22P1-FB2A-DIW1-T01	22P1A UPW/DI MAKE UP TANK	1	SUS 316	15		

UPW/DI PUMP SCHEDULE													
UNIT MARK	SERVICE	TYPE	SUCTION SIZE A	DISCHARGE SIZE A	FLOW L/S	HEAD M	BHP	RPM	MOTOR			FLUID	REMARKS
									POWER (KW)	VOLTAGE	TYPE		
22P1-FB2H-DIW2-P01	22P1B UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横
22P1-FB2H-DIW2-P02	22P1B UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横
22P1-FB2H-DIW2-P03	22P1B UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横(STANDBY)
22P1-FB2A-DIW1-P01	22P1A UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横
22P1-FB2A-DIW1-P02	22P1A UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横
22P1-FB2A-DIW1-P03	22P1A UPW/DI MAKE UP WATER SYSTEM	IN-LINE	100	100	17.5	68	-	3500	19	3/460/60	ODP	DI WATER	W/VFD,IE3 SUS 横(STANDBY)

PLATE TYPE HEAT EXCHANGER SCHEDULE													
UNIT MARK	SERVICE	CAPACITY KW	COLD SIDE					HOT SIDE					REMARKS
			EWI °C	LWI °C	L/S	WTR ΔP KPa	FLUID	EWI °C	LWI °C	L/S	WTR ΔP KPa	FLUID	
22P1-F40H-7CH-HX01	7°C CHILLED WATER SYSTEM	5550	5	13	166.5	<50	WATER	15	7	166.5	<50	WATER	
22P1-F40H-7CH-HX02	7°C CHILLED WATER SYSTEM	5550	5	13	166.5	<50	WATER	15	7	166.5	<50	WATER	
22P1-F40H-7CH-HX03	7°C CHILLED WATER SYSTEM	5550	5	13	166.5	<50	WATER	15	7	166.5	<50	WATER	(FUTURE)
22P1-F40H-33WW-HX01	33°C WARM WATER SYSTEM	1760	27	33	70.4	<50	WATER	35	29	70.4	<50	WATER	
22P1-F40H-33WW-HX02	33°C WARM WATER SYSTEM	1760	27	33	70.4	<50	WATER	35	29	70.4	<50	WATER	(FUTURE)

HVAC PUMP SCHEDULE														
UNIT MARK	SERVICE	TYPE	SUCTION SIZE A	DISCHARGE SIZE A	FLOW L/S	HEAD M	BHP	RPM	MOTOR			FLUID	REMARKS	
									KW	VOLTAGE	TYPE			
22P1-F40H-7CH-P01	7°C CHILLED WATER SYSTEM	IN-LINE	250	250	165	30	-	-	75	3/460/60	TEFC	WATER	W/VFD ,IE3	
22P1-F40H-7CH-P02	7°C CHILLED WATER SYSTEM	IN-LINE	250	250	165	30	-	-	75	3/460/60	TEFC	WATER	W/VFD ,IE3	
22P1-F40H-7CH-P03	7°C CHILLED WATER SYSTEM	IN-LINE	250	250	165	30	-	-	75	3/460/60	TEFC	WATER	W/VFD ,IE3 (STANDBY)	
22P1-F40H-7CH-P04	7°C CHILLED WATER SYSTEM	IN-LINE	250	250	165	30	-	-	75	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)	
22P1-F40H-33WW-P01	33°C WARM WATER SYSTEM	IN-LINE	150	150	70	30	-	-	30	3/460/60	TEFC	WATER	W/VFD ,IE3	
22P1-F40H-33WW-P02	33°C WARM WATER SYSTEM	IN-LINE	150	150	70	30	-	-	30	3/460/60	TEFC	WATER	W/VFD ,IE3 (STANDBY)	
22P1-F40H-33WW-P03	33°C WARM WATER SYSTEM	IN-LINE	150	150	30	30	-	-	30	3/460/60	TEFC	WATER	W/VFD ,IE3 (FUTURE)	

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KEY PLAN

Building Code
A: Office
F: FAB
H: HPM
C: CUP
G: Gas Yard & BSGS
S: Supporting
N: ASC
W: Warehouse

版次	日期	修正者	修正
A	2023/12/27		ISSUE FOR REVIEW

CONTRACTOR

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United Integrated Services

ACCEPTANCE REVIEWED	DATE	ACCEPTANCE REVIEWED	DATE

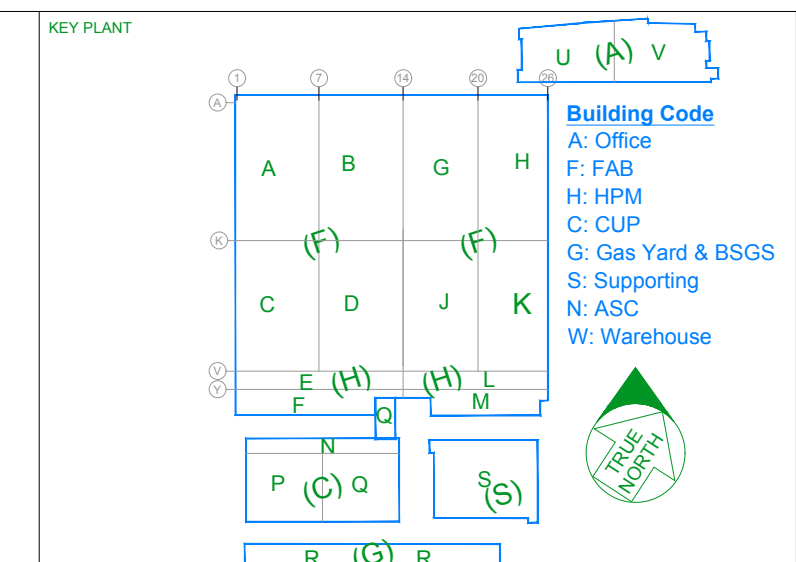
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TITLE

**FLOW DIAGRAM-
PIPING SYSTEM
(8)**

PROJECT	FAB22 Phase 1 PROJECT
PACKAGE NAME	MEP SYSTEM
DRAWING NAME	F22P1-M-2-C-08
SCALE	NOT TO SCALE
FILE NAME	F22P1-M-2-C-08

UNIT	mm
Rev	A

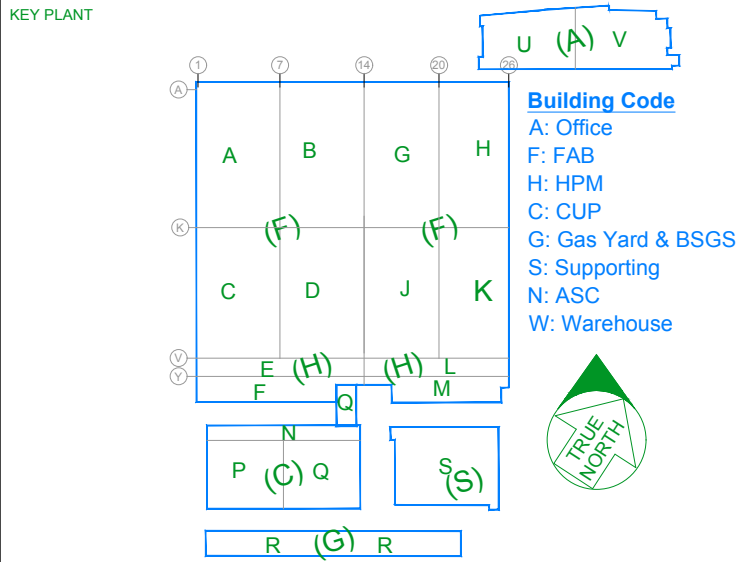
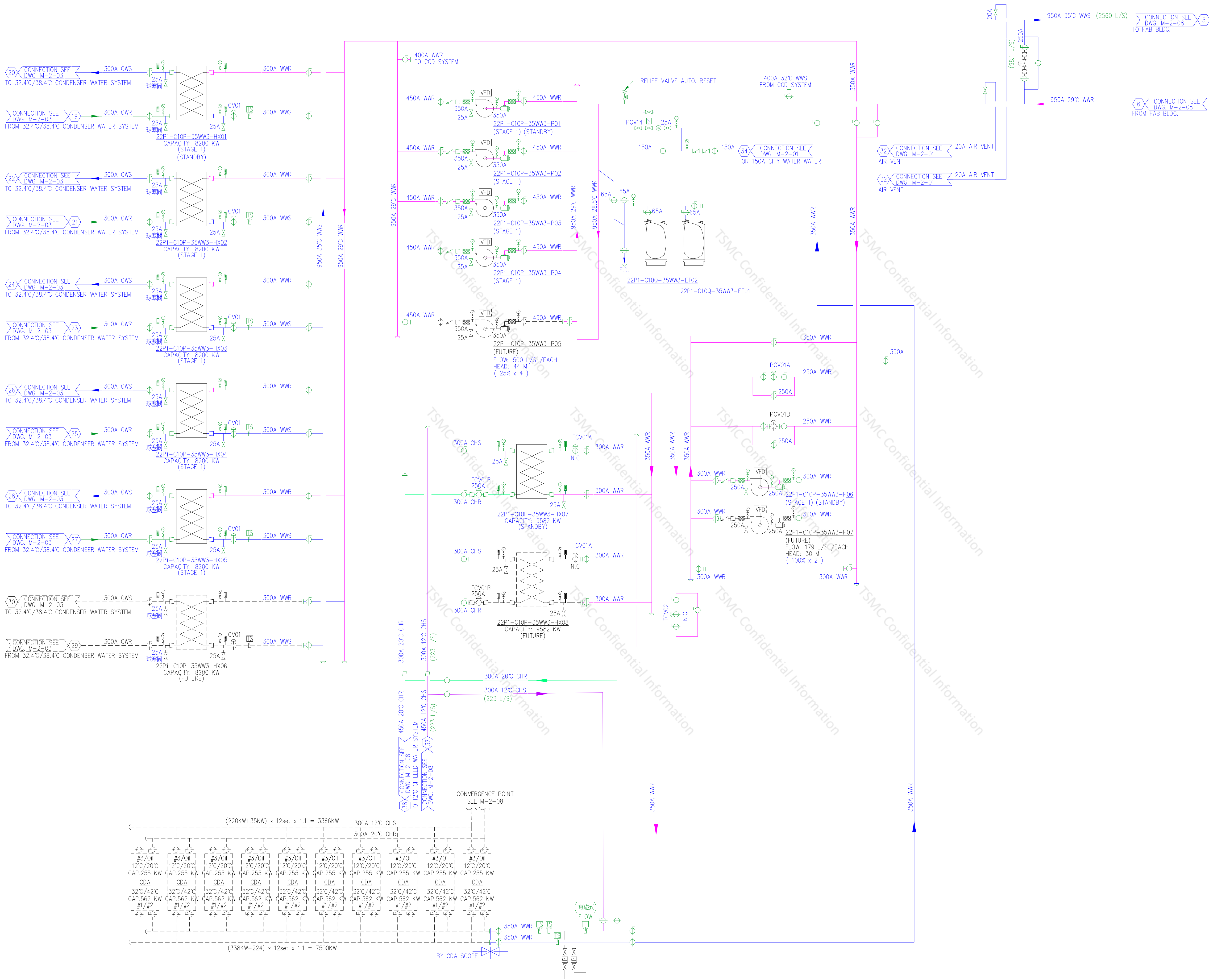


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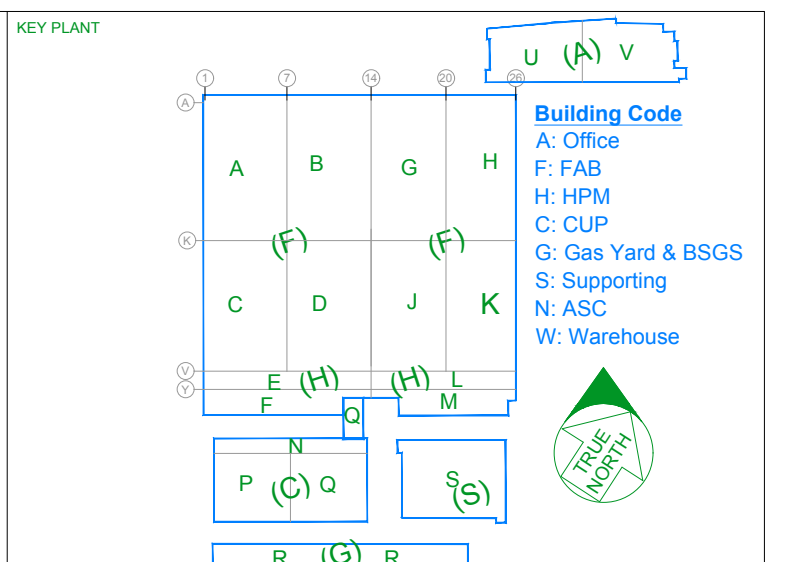
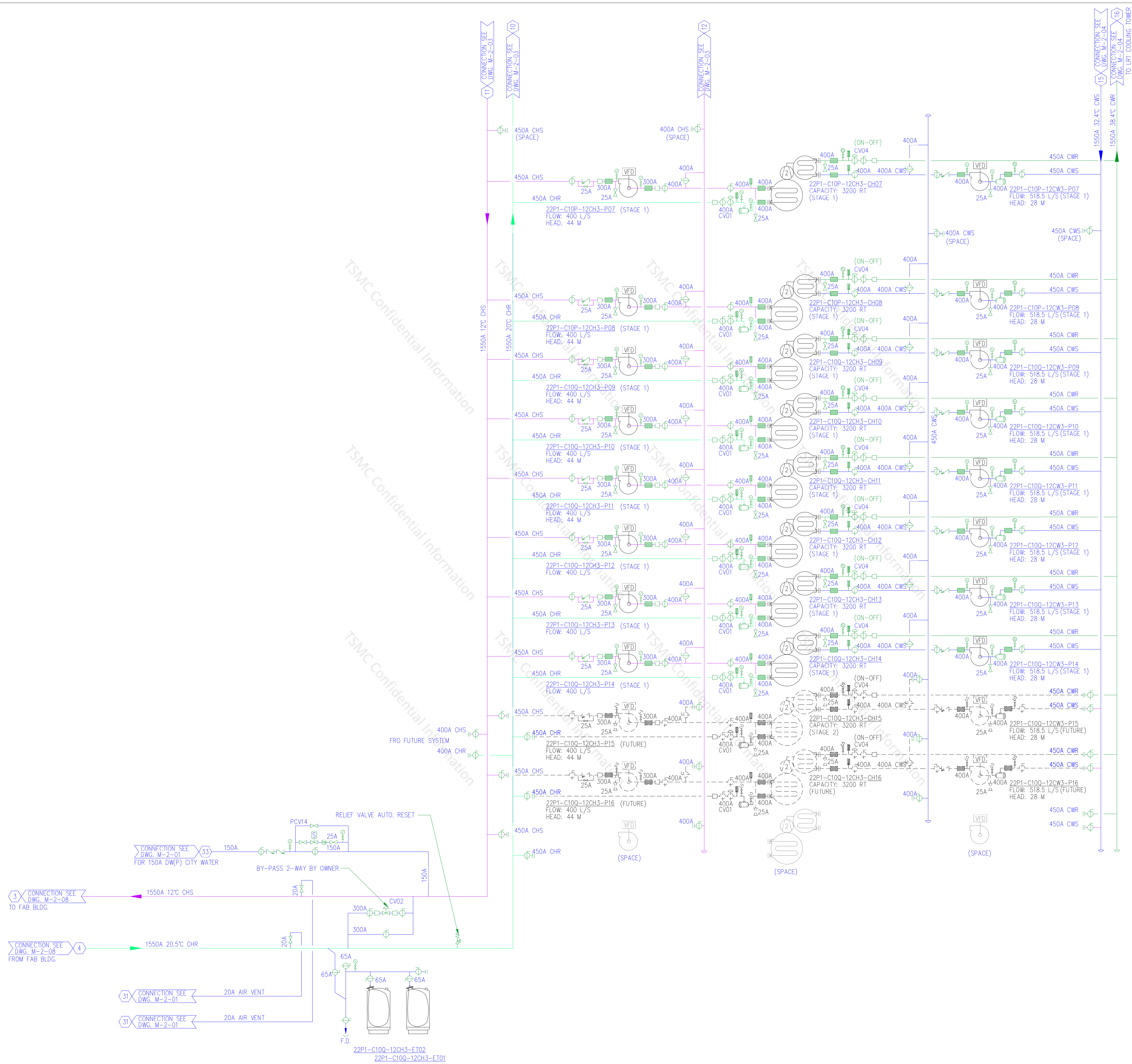
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Taiwan Semiconductor Manufacturing Company, Ltd.

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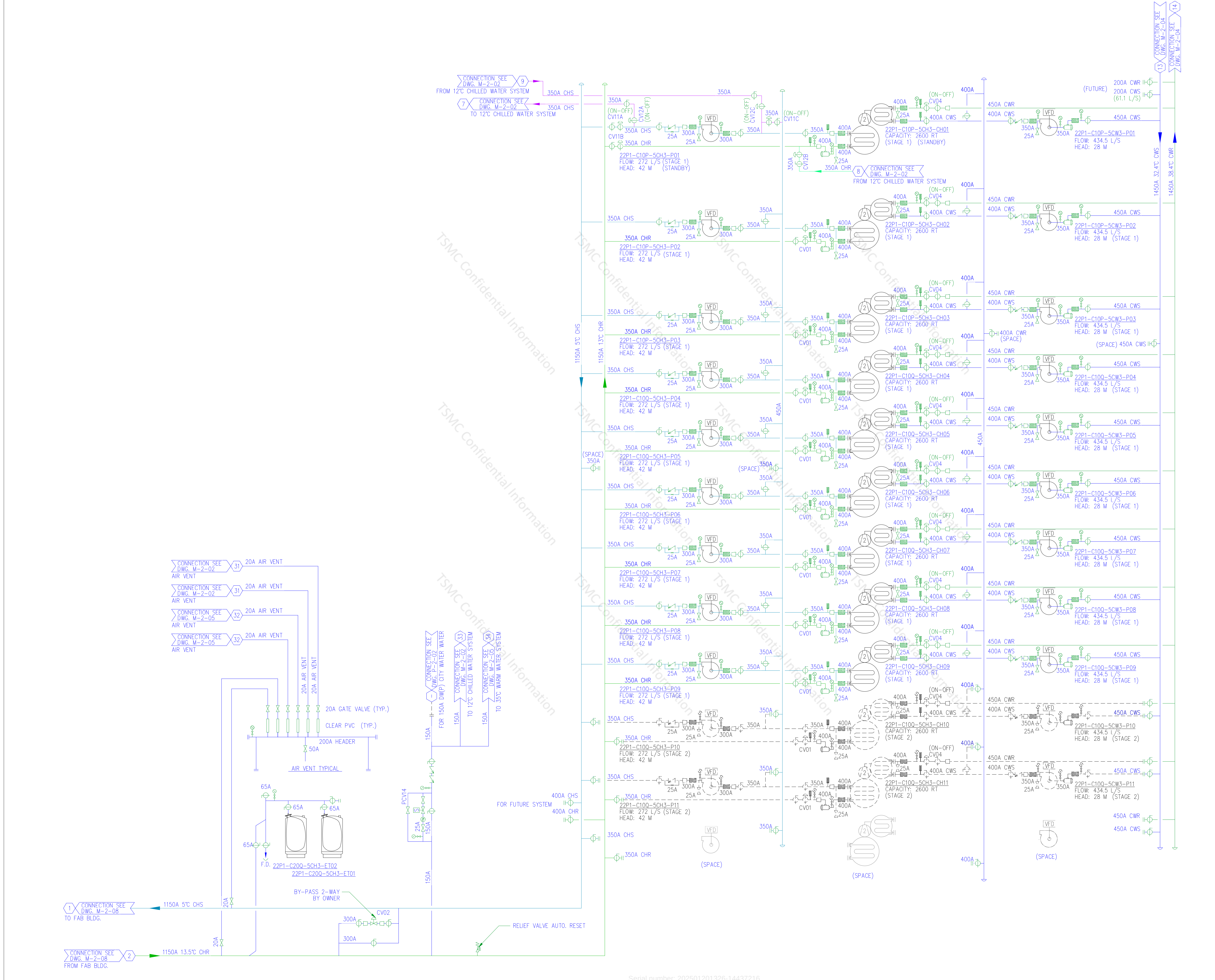


A				2023/11/16	ISSUE FOR REVIEW
版次	日期	修正者	修正	REVISION	

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CONTRACTOR				DATE	DATE	DATE
DESIGNED				DATE	APPROVED	DATE
TITLE				DATE	DATE	DATE
PROJECT				DATE	DATE	DATE
PACKAGE NAME				DATE	DATE	DATE
DRAWING NAME				DATE	DATE	DATE
SCALE				DATE	DATE	DATE
FILE NAME				DATE	DATE	DATE



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KEY PLANT

版次	日期	修正者	修正
VER.	DATE	REVIEW	REVISION
A	2023/11/16		ISSUE FOR REVIEW

台灣積體電路製造股份有限公司
Taiwan Semiconductor Manufacturing Company, Ltd.

ACCEPTANCE REVIEWED	DATE	ACCEPTANCE REVIEWED	DATE
CONTRACTOR		DATE	
漢唐集成股份有限公司 United Integrated Services		DATE	
DRAWN	TCM	DATE	2023/11/16
DESIGNED	TCM	DATE	2023/11/16
TITLE		DATE	
PROJECT		DATE	
PACKAGE NAME		DATE	
DRAWING NAME		DATE	
SCALE		UNIT	
FILE NAME		Rev	

FLOW DIAGRAM
5°C/13°C
CHILLED & CONDENSER
WATER SYSTEM

FAB 22 PHASE 1 PROJECT
MEP SYSTEM
F22P1-M-2-C-01

NOT TO SCALE

2023/11/16